



IMPACT OF SOCIAL ISOLATION ON DEPRESSION AMONG OLDER ADULTS OF INDIA.

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Abstract

India's rapidly ageing population faces heightened depression risks amid shifting family structures and urbanization. This study examines the impact of social isolation on depression among older adults (aged 60+), with a focus on gender differences, using binary logistic regression on data from Wave 1 of the Longitudinal Ageing Study in India (LASI, 2017-18; N=30,741).

Social isolation indicators include living alone, feelings of loneliness, frequency of relative/friend visits, marital status, social organization membership, and grandchildren presence; controls encompass age, education, wealth quintile, residence, and chronic disease. Bivariate analyses reveal strong associations (e.g., chi-square $p < 0.001$ for loneliness, gender). Four regression models confirm robustness: loneliness yields the strongest effect (OR 4.4-9.6, $p < 0.001$), living alone elevates odds (OR 1.23-1.25, $p < 0.001$), and frequent social visits protect (OR 0.64-0.93, $p < 0.05$). Females show 11-14% higher risk (OR 1.11-1.14, $p < 0.05$); higher education and wealth reduce odds (OR 0.73-0.90, $p < 0.01$).

Findings underscore emotional isolation as depression's primary driver, urging policies to enhance social engagement, family support, and targeted interventions for vulnerable women to foster healthy ageing.

(Word count: 198. Keywords: social isolation, depression, older adults, LASI, logistic regression)

Introduction

Population ageing has become a significant demographic trend in India due to improvements in healthcare and living conditions. The growing number of older adults has brought increased attention to issues related to their physical and mental well-being. Among these concerns, depression is one of the most common mental health problems affecting older adults and can significantly reduce their quality of life. In India, rapid socio-economic changes, migration of younger family members, and the gradual decline of traditional joint family systems have contributed to increased vulnerability of older adults to mental health issues.

Social isolation is considered a major factor influencing depression among the elderly. It refers to limited social interaction, lack of emotional support, and reduced participation in social activities. Older adults who live alone, have limited contact with relatives or friends, or feel lonely are more likely to experience depressive symptoms. In the Indian context, changing family structures and urbanization have increased the risk of social isolation among older adults, making it an important area of research.

Gender differences further influence the mental health outcomes of older adults. Female older adults often face greater social and economic disadvantages, which may increase their susceptibility to both social isolation and depression. Despite the growing importance of these issues, there is limited empirical research examining the relationship between social isolation and depression among older adults in India using nationally representative data.

Therefore, this study aims to examine the impact of social isolation on depression among older adults in India and to analyze gender differences in this relationship. Using nationally representative survey data and applying binary logistic regression analysis, the study seeks to provide a better understanding of how social isolation and gender influence depression among the elderly population. The findings are expected to contribute to the development of policies and interventions to improve the mental well-being of older adults in India.

Furthermore, this study contributes to the broader understanding of mental health challenges faced by the ageing population in India by highlighting the importance of social connections and support systems in later life. By providing empirical evidence on the relationship between social isolation, gender, and depression, the study offers valuable insights for researchers, policymakers, and healthcare practitioners to design inclusive social and mental health policies that promote the overall well-being of older adults in the country.

Objective

The present study aims to examine the relationship between social isolation and depression among older adults in India. Specifically, the study seeks to analyze the effect of social isolation on depression and to explore gender differences in depression by assessing whether the impact of social isolation on depression varies between male and female older adults.

Data Source

This study is based on secondary data obtained from the first wave of the Longitudinal Ageing Study in India (LASI) conducted during 2017–18. LASI is a nationally representative survey that provides comprehensive information on the health, social, and economic conditions of individuals aged 45 years and above across all states and union territories of India. For the present study, the analysis is limited to older adults aged 60 years and above. The individual-level data were collected from the official LASI website for academic purposes and include detailed information on depression, social isolation, demographic characteristics, and socio-economic factors, making the dataset suitable for examining the relationship between social isolation and depression among older adults in India.

Variable Description

Response Variable

Depression is the response variable here, And Depression means the common mental health disorder characterised by persistent feelings of sadness, loss of interest or pleasure in activities, low energy, and reduced ability to perform daily functions. It affects emotional, cognitive, and physical well-being and can significantly impair an individual's quality of life, particularly among older adults.

The Question that had asked about Depression in LASI is :

Q. During the last 12 months, was there ever a time when you felt depressed.

1. Rarely or Never
2. Sometimes
3. Often
4. All the time

For the purpose of this study, the variable was recoded into a binary outcome to facilitate logistic regression analysis. Respondents who reported having been diagnosed with depression were coded as “1” (Yes), while those who reported not having depression were coded as “0” (Never). This binary coding of the dependent variable allows for the estimation of the likelihood of depression among older adults in relation to social isolation and other socio-economic factors.

Independent Variables

1. **Living Arrangements:** It refers to the household living situation of older adults, indicating whether they live alone or with family members. In data living arrangement was originally categorized into multiple groups, then the variable is recoded as binary form “1” (living alone) & “0” (living with spouse and/or others).
2. **Marital Status :** Marital status refers to the current marital condition of older adults and serves as an important indicator of social and emotional support. And it also recoded from several groups to 4 groups, married, never married, widow, divorced.
3. **Feels alone :** Its indicating whether an individual frequently feels socially or emotionally isolated. And it is divided in binary groups yes & no.
4. **Visited relative/friends :** indicating the level of social engagement and support among older adults. And it is grouped as Daily, weekly, monthly, yearly & never.
5. **Member of Social organisation/club :** refers to whether an older adult participates in any social, community, or recreational groups, indicating their level of social participation and connectedness. And it has also binary groups yes & no.
6. **Have Grandchildren :** indicating the presence of extended family ties and potential sources of emotional and social support.
7. **Gender :** It shows in which gender the older adults are belongs. And it categorized in two groups male & female.

Control Variables

1. **Age** : it's representing an important demographic factor influencing health and depression among older adults. And here the age started from 60.
2. **Highest level of education** : Indicating their educational attainment and socio-economic status. It has 5 categories.
3. **Wealth-Quantile** : refers to the economic status of respondents based on household consumption or assets, categorized into five groups from poorest to richest, indicating their relative socio-economic position.
4. **Place of residence** : Place of residence refers to the type of area where the respondent lives, categorized as rural or urban, indicating differences in living conditions, access to resources, and social environment.
5. **Chronic disease**: It refers to the presence of long-term health conditions such as hypertension, diabetes, or other persistent illnesses among older adults, which may influence their physical and mental well-being.

Methodology

This study employs a quantitative research design using secondary data from the Longitudinal Ageing Study in India (LASI) Wave 1 (2017–18). The analysis focuses on older adults aged 60 years and above. The primary objective is to examine the effect of social isolation on depression and to assess whether this relationship varies by gender. And the methodologies for this research are as follows:

Descriptive Analysis

The study begins with descriptive statistics to summarize the demographic, socio-economic, and social isolation characteristics of the respondents. Frequency distributions, percentages, and graphical representations are used to describe the prevalence of depression and the distribution of key explanatory variables such as living arrangement, marital status, social participation, loneliness, and other control variables.

Bivariate Analysis

Before estimating the regression models, bivariate analysis is conducted to examine the association between depression and individual explanatory variables. Cross-tabulation and percentage comparisons are used to explore preliminary relationships between depression and social isolation indicators, as well as gender differences. This step helps identify potential associations and guides the specification of the regression model.

Logistic Regression Analysis

To examine the effect of social isolation on depression, binary logistic regression is employed, as the dependent variable (depression) is binary in nature (1 = Yes, 0 = No).

Logistic regression estimates the probability of experiencing depression as a function of social isolation and other socio-economic characteristics.

The basic model is specified as:

$$\text{Logit (Depression)} = \beta_0 + \beta_1(\text{Social Isolation Variables}) + \beta_2(\text{Control Variables}) + \varepsilon$$

Social isolation variables include living arrangement, marital status, membership in social organizations, having grandchildren, visiting relatives or friends, and feelings of loneliness. Control variables include age, gender, education, wealth quintile, chronic disease, and place of residence. To examine gender differences, gender is included as an explanatory variable in the regression model. The estimated coefficients are interpreted in terms of odds ratios, which indicate the likelihood of depression associated with each explanatory variable. Statistical significance is assessed at conventional levels ($p < 0.05$).

All analyses are conducted using Stata and excel software.

Regression Models

To examine how the social isolation effect the depression among older adults, binary logistic regression models are estimated. Since the dependent variable (depression) is dichotomous in nature (1 = Yes, 0 = No), logistic regression is appropriate for estimating the probability of experiencing depression.

The general form of the logistic regression model is expressed as:

$$\text{Log-Odds : } \ln\left(\frac{p}{1-p}\right) = y = \alpha + \beta_1 * x_1 + \beta_2 * x_2 + \dots$$

$$\text{Sigmoid function : } f(x) = \frac{1}{1 + e^{-x}}$$

Here in this research 4 logistic regression model is used. Those are as follows.

Model 1 : Social Isolation & Depression Unadjusted Model

The first model estimates the effect of social isolation variables on depression:

$$\text{Logit}(\text{Depression}_i) = \beta_0 + \beta_1(\text{Living Arrangement}) + \beta_2(\text{Marital Status}) + \beta_3(\text{Feels Alone}) + \beta_4(\text{Visited by friends/relatives}) + \beta_5(\text{Member of organization/club}) + \beta_6(\text{Have Grandchildren}) + \varepsilon_i$$

Model 2 : Social Isolation & Depression adjusted Model

The first model estimates how the effect of social isolation variables on depression changes after adding the socioeconomic variables:

$$\text{Logit}(\text{Depression}_i) = \beta_0 + \beta_1(\text{Living Arrangement}) + \beta_2(\text{Marital Status}) + \beta_3(\text{Feels Alone}) + \beta_4(\text{Visited by friends/relatives}) + \beta_5(\text{Member of organization/club}) + \beta_6(\text{Have GrandChildren}) + \beta_7(\text{Age}) + \beta_8(\text{Wealth Quantile}) + \beta_9(\text{Place of residence}) + \beta_{10}(\text{Education}) + \beta_{11}(\text{Chronic disease}) + \varepsilon_i$$

Model 3 : Gender unadjusted Model

The first model estimates the effect of social isolation variables on depression varies by Sex:

$$\text{Logit}(\text{Depression}_i) = \beta_0 + \beta_1(\text{Living Arrangement}) + \beta_2(\text{Marital Status}) + \beta_3(\text{Feels Alone}) + \beta_4(\text{Visited by friends/relatives}) + \beta_5(\text{Member of organization/club}) + \beta_6(\text{Have Grandchildren}) + \beta_7(\text{Sex}) + \varepsilon_i$$

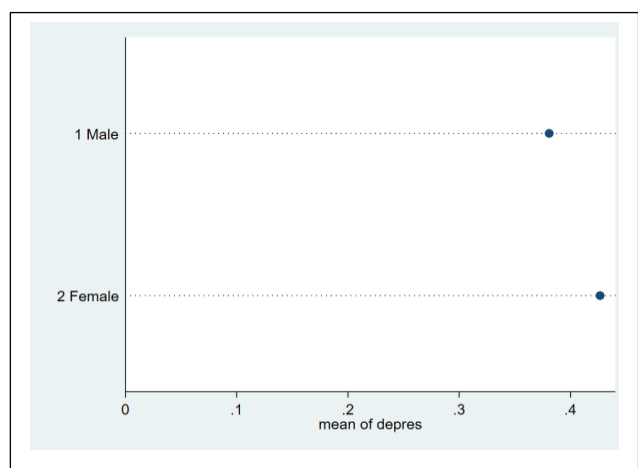
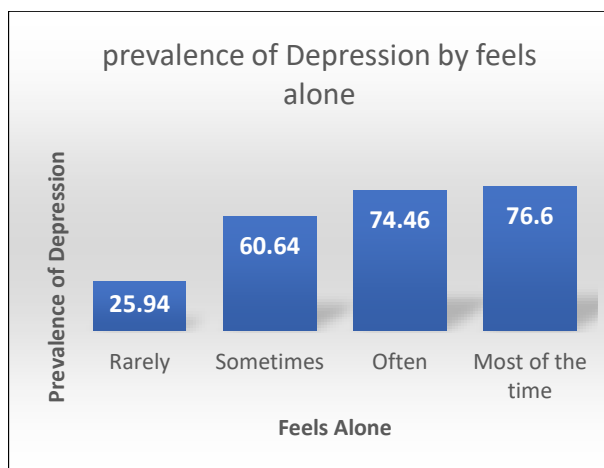
Model 4 : Gender adjusted Model

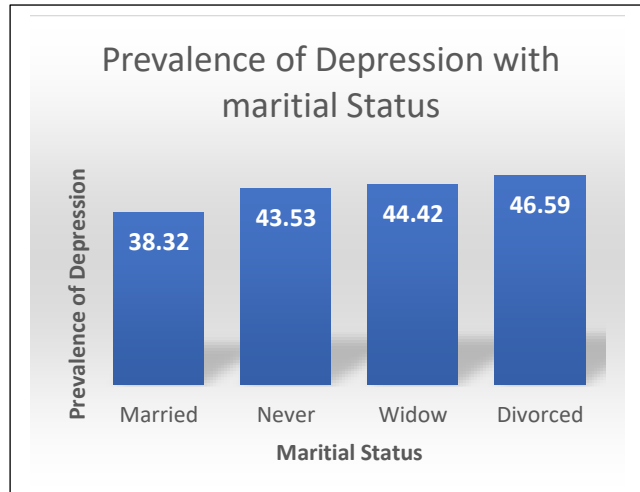
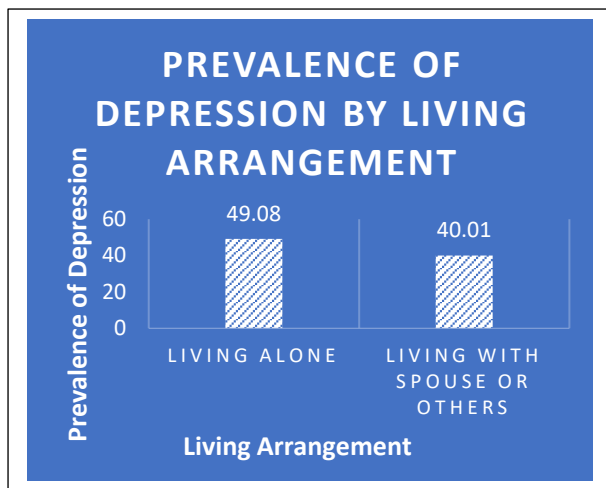
The first model estimates how the effect of social isolation variables on depression varies by sex changes after adding the socioeconomic variables:

$$\text{Logit}(\text{Depression}_i) = \beta_0 + \beta_1(\text{Living Arrangement}) + \beta_2(\text{Marital Status}) + \beta_3(\text{Feels Alone}) + \beta_4(\text{Visited by friends/relatives}) + \beta_5(\text{Member of organization/club}) + \beta_6(\text{Have GrandChildren}) + \beta_7(\text{Sex}) + \beta_7(\text{Age}) + \beta_8(\text{Wealth Quantile}) + \beta_9(\text{Place of residence}) + \beta_{10}(\text{Education}) + \beta_{11}(\text{Chronic disease}) + \varepsilon_i$$

Results and findings

Exploratory Data Analysis





Bivariate Analysis

Bivariate analysis was conducted to examine the association between depression and selected social isolation and socio-economic variables among older adults. Cross-tabulation and percentage distribution were used to understand the preliminary relationship between the dependent and explanatory variables before fitting the binary logistic regression model. This step helps to identify variations in the prevalence of depression across different groups and provides a basis for multivariate analysis. And it helps to detect the important variables and to support the regression results.

The bivariate analysis reveals significant associations between depression and most of the selected social isolation variables among older adults in India. Marital status shows a strong relationship with depression ($\chi^2 = 108.54$, $p < 0.001$), with a higher prevalence observed among widowed (44.42%) and divorced older adults (46.59%) compared to currently married individuals (38.32%). Living arrangement is also significantly associated with depression ($\chi^2 = 51.13$, $p < 0.001$); nearly 49.08% of older adults living alone reported depression compared to 40.01% among those living with others. Feelings of loneliness demonstrate the strongest association ($\chi^2 \approx 5000$, $p < 0.001$), as the prevalence of depression rises sharply from 25.94% among those who never feel alone to 76.60% among those who feel lonely most of the time. Visiting relatives or friends is significantly related to depression ($\chi^2 = 167.16$, $p < 0.001$), with higher prevalence among those who never visit others (46.36%) compared to those with daily interaction (28.90%). Having grandchildren also shows a significant association ($\chi^2 = 11.48$, $p = 0.001$), with slightly higher depression among those without grandchildren (43.39%) than those with grandchildren (40.17%). However, membership in social organizations does not show a statistically significant association with depression ($\chi^2 = 1.58$, $p = 0.209$). Gender differences are evident ($\chi^2 = 66.77$, $p < 0.001$), with females reporting a higher prevalence of depression (42.67%) compared to males (38.09%). Overall, the chi-square results indicate that most indicators of social isolation are significantly associated with depression among older adults, justifying the use of logistic regression to examine the adjusted effects of these variables.

And also Depression was also found to vary across socio-economic characteristics such as age, education, wealth status, and chronic disease.

Logistic Regression Analysis

To examine the impact of social isolation on depression among older adults in India, binary logistic regression analysis was employed, as the dependent variable is dichotomous in nature. The regression analysis was carried out in four stages to understand both the unadjusted and adjusted effects of social isolation on depression and to assess whether these effects vary by gender. The first model includes only social isolation variables to estimate their direct association with depression. The second model incorporates socio-economic and demographic characteristics to examine the net effect of social isolation after controlling for these factors. The third model introduces gender to assess gender differentials in depression, while the final model presents a fully adjusted specification including social isolation, gender, and socio-economic variables simultaneously. The results are presented in terms of odds ratios to facilitate interpretation of the likelihood of depression among older adults.

Table 1 : Social Isolation & Depression Unadjusted Model

depres	Odds ratio	P-Value	95% CI	
mar_st	marital status(ref- Never married)			
Married	0.928269	0.499	0.7480213	1.15195
Widow	0.9888333	0.694	0.9350917	1.045664
Divorced	1.099785	0.688	0.6914774	1.749193
liv_arr	living arrangement(ref- living with spouse or others)			
living Alone	1.232468	0.001	1.094141	1.388284
fs706	feeling alone(ref- Never)			
Sometimes	4.405137	<0.001	4.156482	4.668669
Often	8.370684	<0.001	7.637474	9.174283
Most of the time	9.576755	<0.001	8.370715	10.95656
fs213	Have Grandchildren(ref- yes)			
No	1.067855	0.141	0.9783749	1.165519
vrf_1	visited by friend(ref- never)			
Daily	0.6389349	<0.001	0.5490164	0.7435803
Weekly	0.8462407	0.001	0.7659907	0.9348981
Monthly	0.8799031	<0.001	0.819475	0.9447873
Yearly	0.9211508	0.018	0.8605018	0.9860744
fs501	Member of organization(ref- yes)			
No	0.9880563	0.812	0.8950272	1.090755

Table 2 : Social Isolation & Depression adjusted Model

depres	Odds ratio	P-Value	95% CI	
mar_st	marital status(ref- Never married)			
Married	0.9299365	0.511	0.7486772	1.15508
Widow	0.9732648	0.366	0.9177398	1.032149
Divorced	1.077264	0.754	0.6767079	1.714918
liv_arr	living arrangement(ref- living with spouse or others)			
living Alone	1.243464	<0.001	1.103297	1.401438
fs706	feeling alone(ref- Never)			
Sometimes	4.385995	<0.001	4.137951	4.648906
Often	8.316173	<0.001	7.586658	9.115837
Most of the time	9.518841	<0.001	8.318753	10.89206
fs213	Have Grandchildren(ref- yes)			
No	1.110846	0.021	1.016206	1.2143
vrf_1	visited by friend(ref- never)			
Daily	0.6494153	<0.001	0.557717	0.7561903
Weekly	0.8612325	0.004	0.7790102	0.9521332
Monthly	0.894991	0.003	0.8328024	0.9618234
Yearly	0.9289188	0.035	0.8674327	0.9947632
fs501	Member of organization(ref- yes)			
No	0.9591784	0.414	0.8678922	1.060066
dm005(Age)	0.9991928	0.656	0.995645	1.002753
edu_1	Highest Education(ref- No Education)			
Primary Education	1.002808	0.925	0.9458752	1.063168
Secondary Education	0.8531872	0.003	0.7694277	0.9460647
Higher secondary	0.8040166	0.006	0.6875123	0.9402634
Higher Education	0.7342006	<0.001	0.6420159	0.8396217
mpce_quintile	Wealth Quantile(ref- Poor)			
2 Poorer	0.9004578	0.008	0.8331277	0.9732292
3 Middle	0.9000689	0.008	0.8323327	0.9733176
4 Richer	0.9702408	0.454	0.8963914	1.050174
5 Richest	0.9715606	0.491	0.8950573	1.054603
residence	Place of residence(ref- Rural)			
2 Urban	0.964042	0.205	0.9109702	1.020206
chro_1	Have Chronic disease(ref- yes)			
No	0.8463858	0.001	0.7641177	0.9375112

Table 3 : Gender unadjusted Model

depres	Odds ratio	P-Value	95% CI	
mar_st	marital status(ref- Never married)			
Married	0.9205142	0.452	0.7417768	1.14232
Widow	0.9413837	0.047	0.8869058	0.9992079
Divorced	1.068156	0.781	0.6714623	1.699213
liv_arr	living arrangement(ref- living with spouse or others)			
living Alone	1.082032	<0.001	1.101444	1.397694
fs706	feeling alone(ref- Never)			
Sometimes	4.409577	<0.001	4.160546	4.673514
Often	8.365995	<0.001	7.632981	9.169402
Most of the time	9.612029	<0.001	8.401078	10.99753
fs213	Have Grandchildren(ref- yes)			
No	1.067855	0.141	0.99115	1.181248
vrf_1	visited by friend(ref- never)			
Daily	0.6451324	0.0499585	0.5542843	0.7508707
Weekly	0.846336	0.0430417	0.766044	0.9350438
Monthly	0.8797143	0.0319418	0.8192853	0.9446005
Yearly	0.9220732	0.0320502	0.861348	0.9870796
fs501	Member of organization(ref- yes)			
No	1.082032	0.0484332	0.8857844	1.079863
dm003	Sex of Respondents(ref- Male)			
female	1.136534	0.0313095	1.076795	1.199586

Table 4 : Gender adjusted Model(final Model)

depres	Odds ratio	P-Value	95% CI	
mar_st	marital status(ref- Never married)			
Married	0.9221298	0.464	0.7424199	1.14534
Widow	0.9328502	0.029	0.8764918	0.9928324
Divorced	1.051065	0.834	0.6600915	1.673612
liv_arr	living arrangement(ref- living with spouse or others)			
living Alone	1.249075	<0.001	1.108231	1.407819
fs706	feeling alone(ref- Never)			
Sometimes	4.393982	<0.001	4.145355	4.657521
Often	8.321269	<0.001	7.59113	9.121636
Most of the time	9.554796	<0.001	8.349728	10.93378
fs213	Have Grandchildren(ref- yes)			
No	1.123754	0.01	1.027809	1.228655
vrf_1	visited by friend(ref- never)			
Daily	0.6537268	<0.001	0.5613716	0.7612761
Weekly	0.8605291	0.003	0.7783544	0.9513793
Monthly	0.8949309	0.003	0.8327332	0.9617742
Yearly	0.9299905	0.038	0.8684185	0.9959282
fs501	Member of organization(ref- yes)			
No	0.9568243	0.387	0.8657329	1.0575
dm005(Age)	1.000388	0.833	0.996787	1.004001
edu_1	Highest Education(ref- No Education)			
Primary Education	1.035675	0.257	0.9748145	1.100336
Secondary Education	0.8933251	0.037	0.8036457	0.9930119
Higher secondary	0.8421428	0.033	0.7189088	0.9865015
Higher Education	0.7727315	<0.001	0.6741102	0.8857809
mpce_quintile	Wealth Quantile(ref- Poor)			
2 Poorer	0.8977648	0.007	0.8305976	0.9703634
3 Middle	0.8946749	0.005	0.8272729	0.9675684
4 Richer	0.9631778	0.354	0.8897736	1.042638
5 Richest	0.9601768	0.333	0.884372	1.042479
residence	Place of residence(ref- Rural)			
2 Urban	0.9497777	0.077	0.8970445	1.005611
chro_1	Have Chronic disease(ref- yes)			
No	0.8405073	0.001	0.7587552	0.9310678
dm003	Sex of Respondents(ref- Male)			
female	1.115117	<0.001	1.062039	1.191941

Interpretation of Logistic Models

Binary logistic regression analysis was carried out to examine the impact of social isolation on depression among older adults in India and to assess gender differences in this relationship. Four models were estimated, including unadjusted, socio-economic adjusted, gender unadjusted, and fully adjusted models. The results across all models consistently indicate that social isolation plays a crucial role in determining depression among older adults.

The findings reveal that feelings of loneliness emerge as the strongest and most consistent predictor of depression. Compared to older adults who never feel lonely, those who sometimes feel lonely are about 4.4 times more likely to experience depression ($OR \approx 4.40$, $p < 0.001$). The likelihood increases substantially among those who often feel lonely, who are about 8.3 times more likely to suffer from depression ($OR \approx 8.32$ – 8.37 , $p < 0.001$). Furthermore, older adults who feel lonely most of the time are nearly 9.5 times more likely to experience depression ($OR \approx 9.51$ – 9.61 , $p < 0.001$). This strong and statistically significant association remains robust across all four models, even after controlling for gender and socio-economic characteristics, indicating that emotional isolation is the most important determinant of depression among older adults.

Living arrangement also shows a significant association with depression. Older adults living alone have higher odds of depression compared to those living with spouse or others. In the unadjusted model, the odds of depression among those living alone are about 1.23 times higher ($OR = 1.23$, $p = 0.001$). This relationship remains statistically significant in the adjusted and final models ($OR \approx 1.24$ – 1.25 , $p < 0.001$), suggesting that lack of co-residential support increases vulnerability to depression in later life.

Social interaction with relatives and friends demonstrates a protective effect against depression. Compared to older adults who never visit relatives or friends, those who interact daily show about 35–36% lower likelihood of depression ($OR \approx 0.64$, $p < 0.001$). Similarly, weekly ($OR \approx 0.85$, $p < 0.01$), monthly ($OR \approx 0.88$ – 0.89 , $p < 0.01$), and yearly visits ($OR \approx 0.92$ – 0.93 , $p < 0.05$) significantly reduce the likelihood of depression. These findings highlight the importance of regular social contact in maintaining mental well-being among older adults.

Marital status does not exhibit a significant association with depression in most regression models after controlling for other variables, although bivariate results indicated some differences. This suggests that the effect of marital status on depression is largely mediated by factors such as loneliness and social interaction. Having grandchildren also shows limited or inconsistent significance across models, indicating that the mere presence of extended family may not necessarily protect against depression unless accompanied by meaningful interaction.

Gender differences are evident in both gender-unadjusted and fully adjusted models. Female older adults have significantly higher odds of depression compared to males, with approximately 11–14% higher likelihood of experiencing depression ($OR \approx 1.11$ – 1.14 , $p < 0.05$). This indicates greater vulnerability of women to depression in older age. Among socio-economic factors, higher levels of education are associated with lower likelihood of depression. Older adults with higher education are about 20–27% less likely to experience depression compared to those with no education ($OR \approx 0.73$ – 0.84 , $p < 0.01$).

Similarly, better economic status and absence of chronic disease are associated with lower odds of depression, indicating the importance of health and socio-economic well-being.

Overall, the results clearly demonstrate that emotional isolation, particularly loneliness, is the most powerful predictor of depression, followed by living arrangement and frequency of social interaction. The consistency of these findings across all four models confirms the robustness of the relationship between social isolation and depression among older adults in India. These results highlight the critical importance of strengthening social support systems and promoting social engagement to improve mental health outcomes among the ageing population.

Comparison of the models and goodness of the fit

Model	Description	LR Chi ²	Pseudo R ²	Log Likelihood	N	Interpretation
Model 1	Social Isolation (Unadjusted)	5086.12	0.1226	-18201.273	30,741	Baseline model showing crude effect of isolation variables
Model 2	Social Isolation + Socio-economic (Adjusted)	5151.74	0.1242	-18168.466	30,741	Improved fit after adding socio-economic controls
Model 3	Gender Unadjusted Model	5107.70	0.1231	-18190.483	30,741	Shows effect of gender without socio-economic controls
Model 4	Fully Adjusted Model (Isolation + Gender + Socio-economic)	5167.78	0.1246	-18160.448	30,741	Best fitting and most comprehensive model

The comparison of the four logistic regression models indicates a gradual improvement in model fit with the inclusion of additional explanatory variables. Model 1, which includes only social isolation variables, explains about 12.26% of the variation in depression (Pseudo R² = 0.1226) and serves as the baseline model. With the inclusion of socio-economic variables in Model 2, the LR chi-square value increases from 5086.12 to 5151.74 and the Pseudo R² rises to 0.1242, indicating an improvement in explanatory power and overall model performance.

Model 3 incorporates gender without socio-economic controls and shows a slight improvement over the unadjusted model, with an LR chi-square of 5107.70 and Pseudo R² of 0.1231. However, the most comprehensive and best-fitting model is Model 4, which includes social isolation, gender, and socio-economic variables simultaneously. This model reports the highest LR chi-square value (5167.78), highest Pseudo R² (0.1246), and the least negative log likelihood (-18160.448), indicating the best goodness of fit among all models. Since the number of observations remains constant across models (N = 30,741), the comparison is reliable. Overall, the results suggest that the fully adjusted model provides the most robust and comprehensive explanation of depression among older adults in India.

Key Findings

The findings of the study clearly demonstrate that social isolation is a major determinant of depression among older adults in India. The regression results reveal that loneliness is the most powerful predictor of depression, with older adults who sometimes feel lonely being more than four times, and those who often or always feel lonely being nearly eight to ten times more likely to suffer from depression compared to those who never feel lonely. Older adults living alone also show significantly higher odds of depression, highlighting the importance of co-residential and emotional support in later life. In contrast, frequent social interaction with relatives and friends significantly reduces the likelihood of depression, indicating the protective role of social engagement. The analysis further shows that female older adults are more vulnerable to depression than males, and that higher education and better economic status are associated with lower risk of depression, while the presence of chronic disease increases vulnerability. The consistency of these results across all regression models confirms the robustness of the relationship between social isolation and depression. Overall, the study strongly establishes that emotional and social connectedness play a crucial role in shaping the mental health of older adults, emphasizing the need for policies and interventions aimed at reducing loneliness and strengthening social support systems for the ageing population in India.

Conclusion

This study furnishes robust empirical evidence on the association between social isolation and depression among older adults in India, drawing from nationally representative data and rigorous statistical methods. The results unequivocally establish social isolation—particularly emotional isolation—as a pivotal driver of depression. Loneliness exhibits the strongest and most consistent association with depressive symptoms across all regression models, underscoring the profound psychological toll of perceived social disconnection in later life. Older adults living alone or with limited contact from relatives and friends face markedly elevated depression risk, whereas frequent social engagement serves as a potent protective factor.

The analysis further illuminates salient gender and socioeconomic disparities. Women evince greater depression vulnerability than men, attributable to lifelong accumulations of social, economic, and health disadvantages. Higher educational attainment and economic status correlate with reduced depression odds, while chronic morbidity substantially elevates risk, affirming the interplay of social and health determinants in shaping late-life mental health. The persistence of these patterns across models attests to the findings' robustness.

In sum, these results spotlight social isolation as a pressing public health priority amid India's demographic ageing. Bolstering family and community networks, fostering social participation, and tailoring mental health interventions for older adults are imperative to enhance psychological well-being and quality of life. Policies targeting loneliness and social connectedness hold substantial promise for curbing depression and promoting healthy ageing in this burgeoning elderly population.

Policy Implication

India's older adults face rising depression risks from loneliness and isolation, as shown by LASI data analysis. Governments and communities must prioritize simple, actionable steps to build social ties and boost mental health.

Key Recommendations

- Launch nationwide "Senior Buddy" programs pairing isolated elders with volunteers for weekly calls or visits, targeting those living alone (23% higher depression odds).
- Integrate free mental health screenings into routine health check-ups, focusing on women (11-14% higher risk) and those with chronic diseases.
- Promote community centers with daily activities like group exercises or hobby clubs to encourage regular friend/family interactions (35% depression risk reduction).
- Offer tax incentives for families or employers providing elder care support, countering urban migration's isolation effects.
- Educate schools and workplaces on "adopt-a-grandparent" initiatives to foster intergenerational bonds and reduce loneliness (strongest predictor, up to 9.6x risk).

These steps are cost-effective, scalable, and backed by evidence—strengthening families and neighbourhoods for healthier ageing.

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